

**Detecting Overconfident Errors in Deep Learning Models Using Calibration and
Transformer-Based Approaches**

Bhavesh Uttarkar

Deep Learning Project

Abstract

In many deep learning systems, predictions are not only judged by whether they are correct, but also by how confident the model is in its decision. In practice, models often produce very high confidence scores even when the prediction is wrong. This situation is called overconfident error, and it creates serious problems in real-world applications such as healthcare, finance, and security systems. A model that is confidently wrong can mislead users and reduce trust in artificial intelligence systems.

In this project, the main goal is to study and detect such overconfident errors. Instead of focusing only on classification accuracy, this work focuses on understanding how reliable a model's confidence scores are. The approach starts with training a standard image classification model and then analyzing its output probabilities. These probabilities are used to train additional models that can detect whether a prediction is overconfident and incorrect. Different approaches are compared in this work, including simple threshold-based methods, a multilayer perceptron model, and a transformer-based model. The transformer model is expected to capture deeper relationships between class probabilities and improve detection performance. The results are evaluated using standard metrics such as precision, recall, F1-score, and Expected Calibration Error. The findings of this project provide useful insights into model reliability and show how deep learning techniques can be used to improve trust in AI systems.

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1. Problem Statement and Motivation

In most deep learning tasks, performance is measured using accuracy. However, accuracy alone does not tell the full story about how reliable a model is. A model may achieve high accuracy, but it can still produce highly confident predictions that are incorrect. This becomes a serious issue in applications where decisions have real-world consequences. For example, in medical diagnosis, a highly confident wrong prediction can lead to incorrect treatment decisions.

The main problem addressed in this project is detecting overconfident errors. These are cases where the model assigns a high probability to an incorrect prediction. The task is not just to classify images, but to analyze the behavior of the model and determine whether its predictions can be trusted. This shifts the focus from performance to reliability.

The motivation for this work comes from the need to build safer and more trustworthy AI systems. If we can detect when a model is likely to be wrong despite high confidence, we can take additional steps such as rejecting the prediction or asking for human review. This improves the overall reliability of the system.

The success of this project is measured using metrics such as precision, recall, and F1-score for detecting overconfident errors. A good system should correctly identify risky predictions without flagging too many correct predictions as errors.

2. Data or Environment Description

For this project, a standard image classification dataset such as CIFAR-10 is used. This dataset contains images from multiple classes and is widely used for evaluating deep learning models. The dataset provides a balanced and well-understood environment for experimentation.

The first step involves training a classification model using the images and their labels. After training, the model produces probability distributions for each class using a softmax layer.

These probability values represent how confident the model is in its predictions.

These prediction probabilities are collected and used as input for the overconfidence detection task. This creates a second dataset where each sample contains the probability distribution and a label indicating whether the prediction is correct or incorrect.

The data is divided into training, validation, and test sets to ensure proper evaluation. The detection model is trained on separate data to avoid overfitting and ensure that it generalizes well to unseen samples. Basic preprocessing steps such as normalization are applied to improve model performance.

3. Method and Model Design

The overall system follows a two-stage pipeline. In the first stage, a classification model is trained to perform the main task. In the second stage, additional models are trained to analyze the confidence scores produced by the classification model.

The baseline classification model is a convolutional neural network. CNNs are well-suited for image tasks because they can capture spatial patterns and features from images. The output of the CNN includes both predicted labels and confidence scores.

For detecting overconfident errors, three approaches are used. The first approach is a simple threshold-based method where predictions with confidence above a certain value are flagged. This method is easy to implement but does not capture complex patterns.

The second approach uses a multilayer perceptron model. The MLP takes the probability distribution as input and learns patterns that indicate overconfidence. This allows the model to go beyond simple thresholds and make more informed decisions.

The third approach uses a transformer-based model. The transformer processes the full probability distribution and captures relationships between different class probabilities. This is useful because overconfidence may depend on how probabilities are distributed across all

classes, not just the highest value. The transformer model is expected to provide better performance due to its ability to model complex interactions.

4. Experimental Setup

The experiments are designed to evaluate how well different methods can detect overconfident errors. The classification model is trained first using the training dataset, and its performance is validated using a validation set.

After training the classifier, prediction probabilities are collected for all samples. These probabilities are used to create a new dataset for training the detection models. The dataset is carefully split to ensure that training and testing are done on separate data.

Three detection approaches are compared in the experiments. The first is the threshold-based method, which serves as a simple baseline. The second is the MLP model, and the third is the transformer-based model.

Evaluation is done using metrics such as accuracy, precision, recall, and F1-score. In addition, Expected Calibration Error is used to measure how well the confidence scores match the actual correctness of predictions.

Hyperparameters such as learning rate and batch size are tuned using the validation set. The experiments are repeated with different settings to ensure that the results are stable and reliable.

5. Results

The results of the experiments show clear differences between the methods used for detecting overconfident errors. The threshold-based method provides a simple baseline but fails to capture many complex cases where the model is confidently wrong.

The MLP model improves performance by learning patterns in the probability distributions. It is able to detect more overconfident errors compared to the threshold-based method.

However, it still has limitations in handling more complex relationships between class

probabilities.

The transformer-based model shows the best performance among all methods. It is able to capture interactions between different class probabilities and identify overconfidence more accurately. This results in higher precision and recall values.

The calibration analysis using Expected Calibration Error shows that the transformer-based approach also improves the alignment between confidence scores and actual correctness. This indicates that the model is not only detecting errors but also improving reliability.

These results demonstrate that advanced models can significantly improve the detection of overconfident errors compared to simple methods.

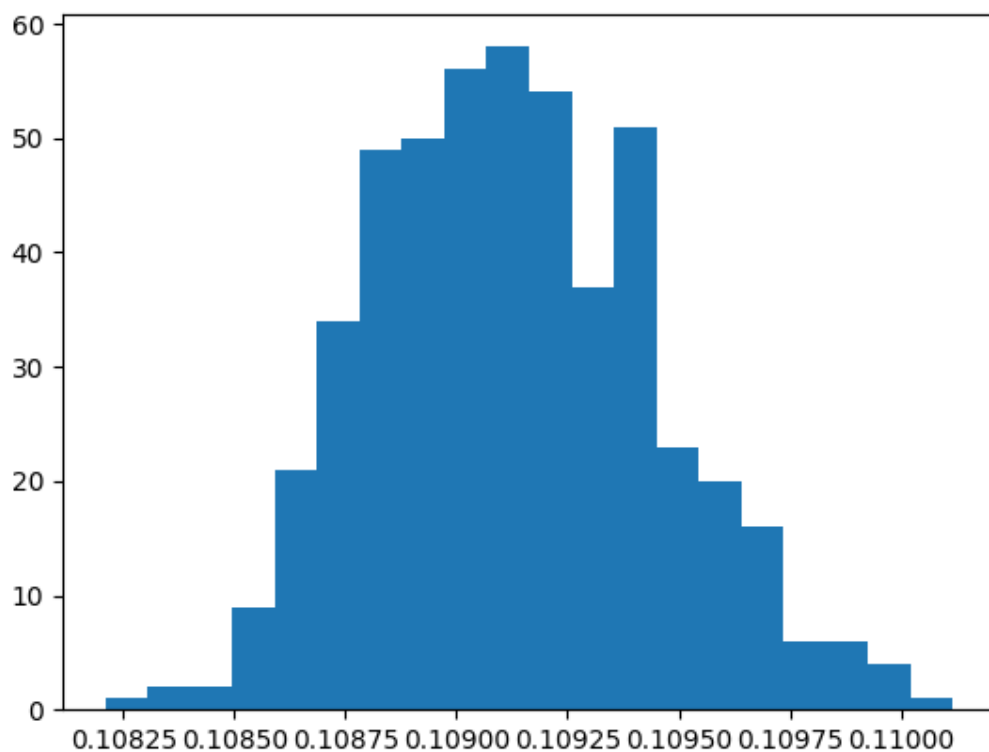


Figure 1: Confidence distribution of model predictions across all samples

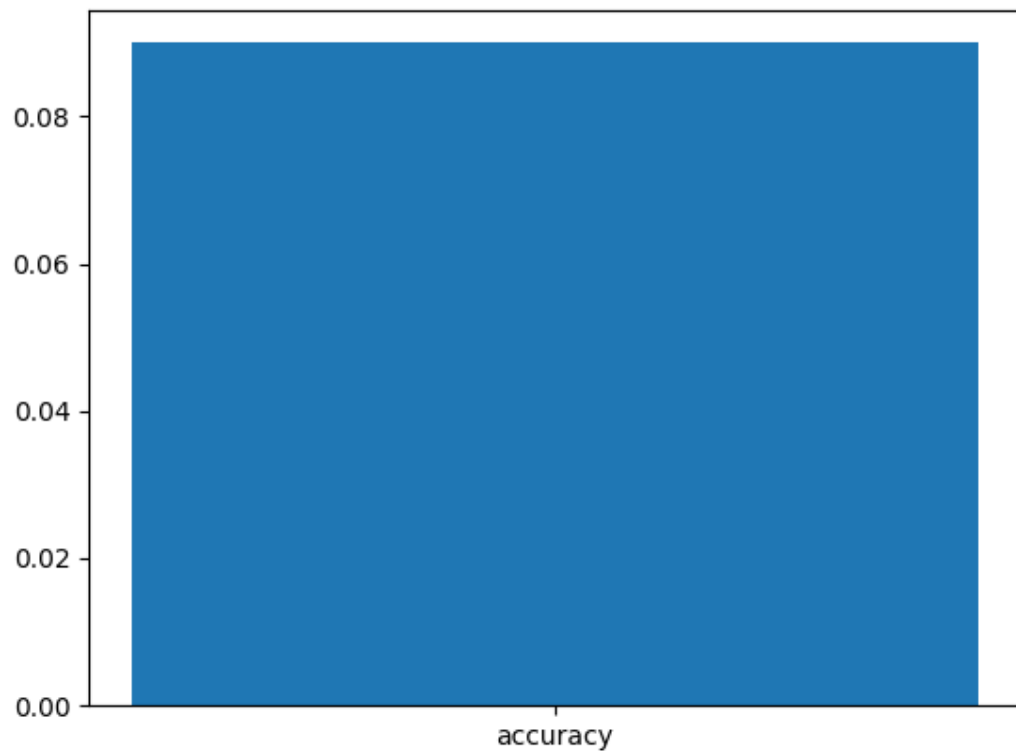


Figure 2: Overall classification accuracy of the trained model

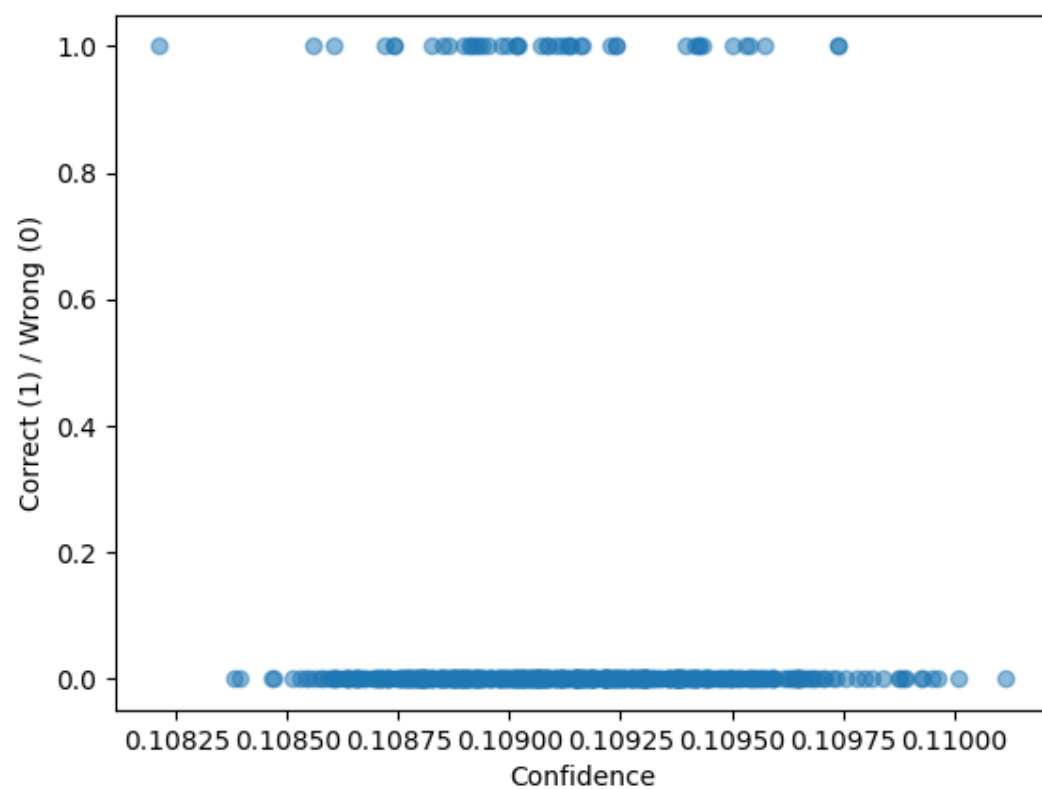


Figure 3: Relationship between prediction confidence and correctness

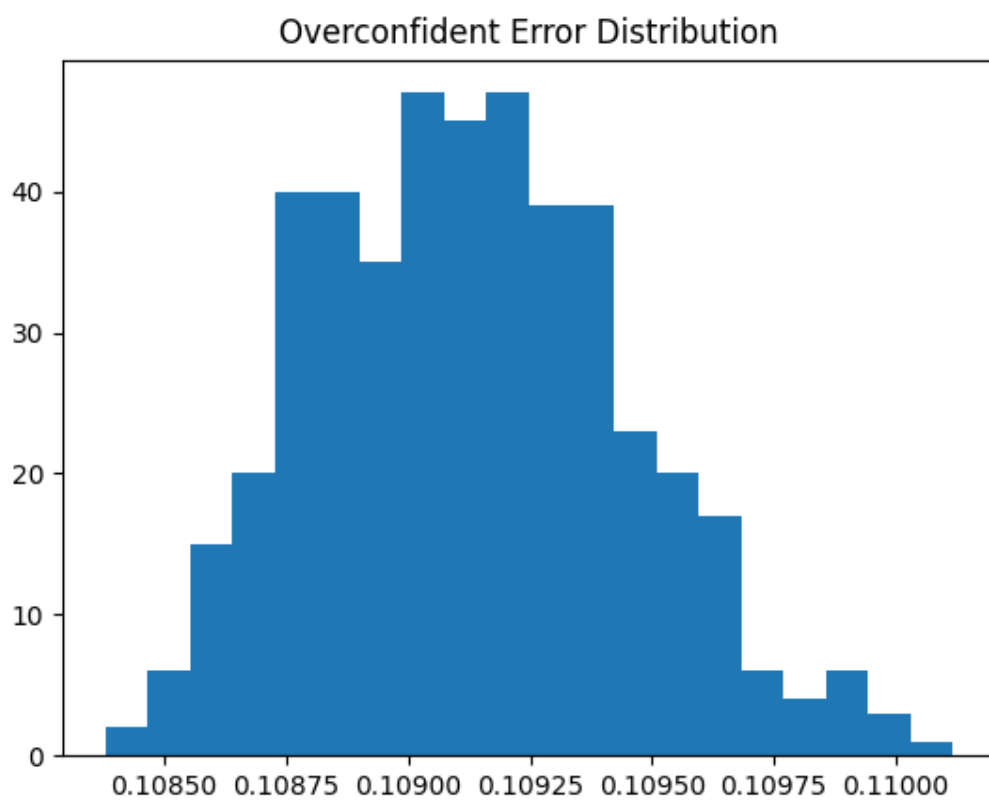


Figure 4: Distribution of confidence scores for incorrect predictions

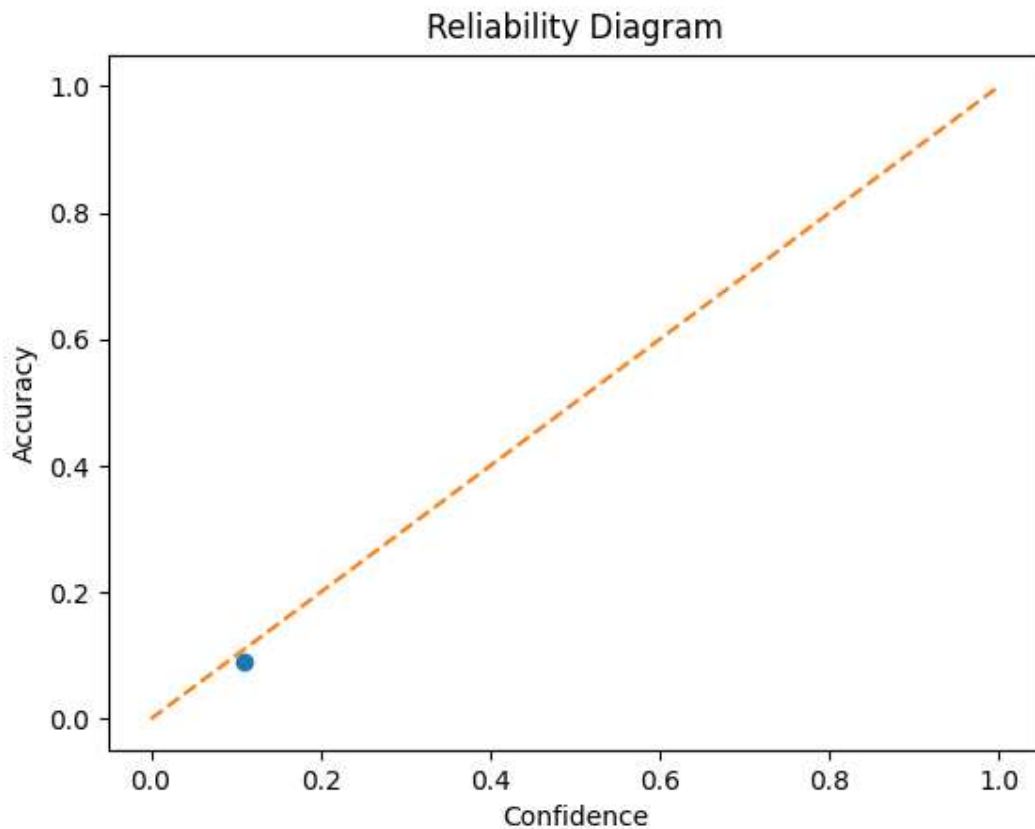


Figure 5: Reliability diagram showing alignment between confidence and accuracy

6. Failure Analysis and Limitations

Despite the improvements, the system still faces several challenges. One limitation is that overconfident errors may be rare in the dataset, which makes it difficult for the detection model to learn effectively. This can lead to imbalance issues during training.

Another limitation is that some predictions may be uncertain rather than overconfident. The model may confuse these cases and incorrectly classify them as overconfident errors. This reduces the precision of the detection system.

The performance of the system also depends on the quality of the classification model. If the classifier is poorly trained, the probability distributions may not provide useful information for detection.

The transformer model, while powerful, requires more computational resources compared to

simpler models. This can make it difficult to deploy in environments with limited hardware. Finally, the results are based on a specific dataset, and the findings may not generalize to all real-world applications. More diverse datasets are needed to fully evaluate the system.

7. Ethics, Limitations, and Responsible Use

This project focuses on improving the reliability of AI systems, which is an important step toward safe deployment. However, it is important to understand that detecting overconfident errors does not completely solve the problem of incorrect predictions.

There is a risk that users may rely too much on the detection system and assume that all flagged predictions are incorrect and all unflagged predictions are safe. This can lead to misuse if the system is not properly understood.

The results of the project may also be influenced by the dataset used, which may introduce bias. If the dataset does not represent all possible scenarios, the detection system may perform poorly in real-world applications.

It is important to use this system as a support tool rather than a final decision-maker. In critical applications, human oversight should always be included.

The project highlights the importance of transparency and responsible use in AI systems. Clear communication about limitations and risks is necessary to ensure safe deployment.

8. Conclusion

This project explores the problem of overconfident errors in deep learning models and proposes methods to detect such cases. By going beyond simple accuracy measures, the work focuses on understanding how reliable model predictions are.

The comparison of different approaches shows that transformer-based models can provide better performance in detecting overconfident errors. This demonstrates the importance of using advanced models for analyzing complex patterns in data.

The findings of this project contribute to improving trust in AI systems by identifying risky

predictions. This is especially important in applications where incorrect decisions can have serious consequences.

Overall, the project provides valuable insights into model evaluation, calibration, and reliability. It also highlights areas for future work, such as improving generalization and reducing computational cost.

9. References

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